

ESE-2025



PRELIMINARY EXAMINATION YOUTUBE SERIES

Digital Electronics

Full Syllabus



Lecture No. 01

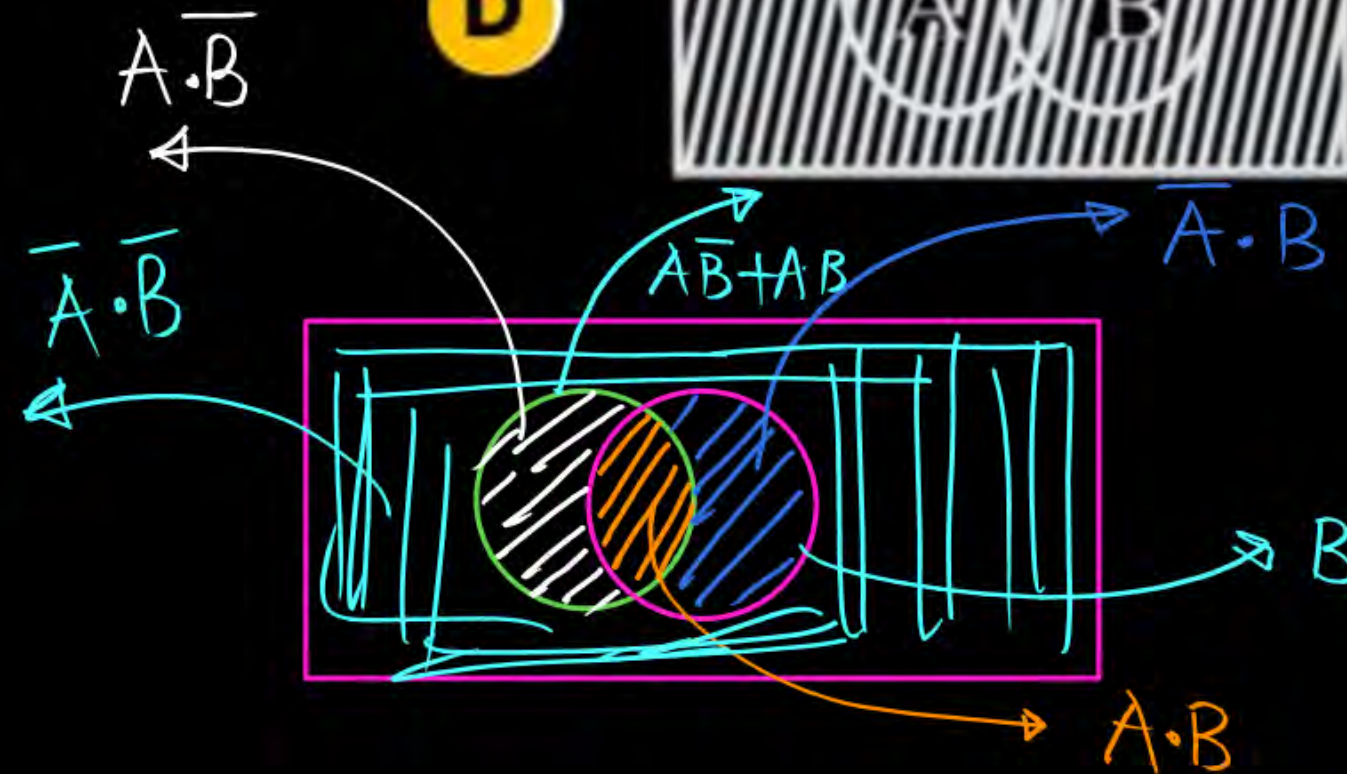
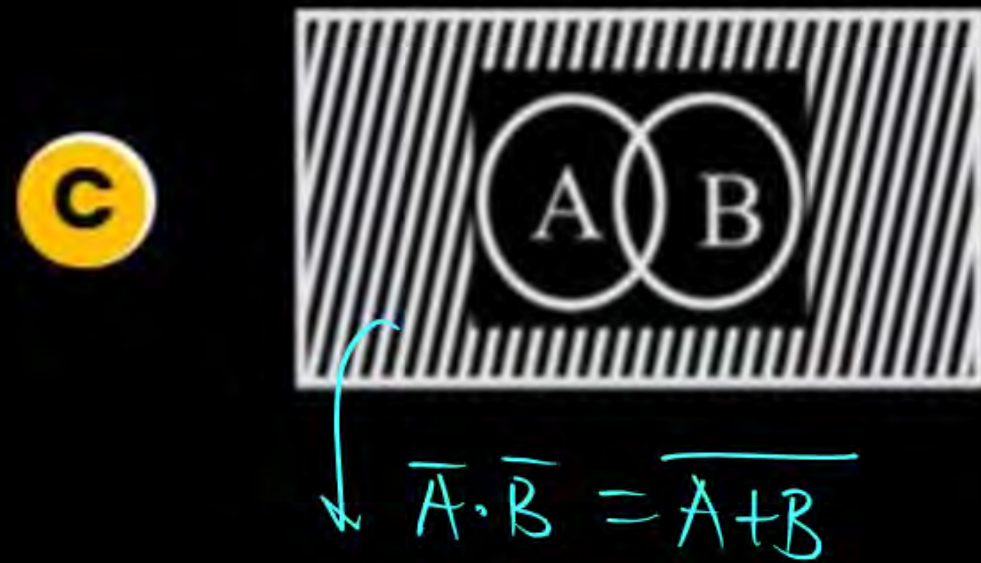
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Topics to be covered

Question Practice

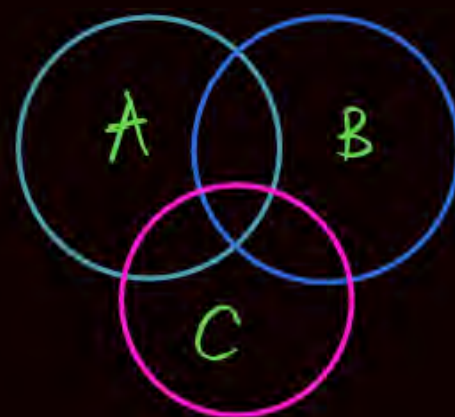


#Q. The Venn diagram representing the Boolean expression $A + (\bar{A} \cdot B)$ is



$$A + (\bar{A} \cdot B) = (A + B)$$

$$= A\bar{B} + AB + \bar{A}B$$



#Q. Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I		List-II	
A.	$A \oplus B = 0$	1.	$A \neq B$
B.	$\overline{A + B} = 0$	2.	$A = B$
C.	$\overline{A} \cdot B = 0$	3.	$A = 1 \text{ or } B = 1$
D.	$A \oplus B = 1$	4.	$A = 1 \text{ or } B = 0$

Handwritten notes and arrows:
 - An arrow points from $A \oplus B = 0$ to $A = B$ (List-II item 2).
 - An arrow points from $\overline{A} \cdot B = 0$ to $\overline{A} = 0 \text{ or } B = 0$ and $A = 1 \text{ or } B = 0$ (List-II item 4).
 - Another arrow points from $\overline{A} \cdot B = 0$ to $A = 1 \text{ or } B = 0$ (List-II item 4).

~~A~~ $A \rightarrow 3, B \rightarrow 2, C \rightarrow 1, D \rightarrow 4$

☒ B $A \rightarrow 2, B \rightarrow 3, C \rightarrow 4, D \rightarrow 1$

~~C~~ $A \rightarrow 3, B \rightarrow 2, C \rightarrow 4, D \rightarrow 1$

~~D~~ $A \rightarrow 2, B \rightarrow 3, C \rightarrow 1, D \rightarrow 4$

#Q. Consider the following statements:

1. Minimization using Karnaugh map may not provide unique solution. ✓
2. Redundant grouping in Karnaugh map may result in non-minimized solution. ✓
3. Don't care states if used in Karnaugh map for minimization, the minimal solution is not obtained. ✗

Which of the statements given above are correct?

- A** 1, 2 and 3
- B** 2 and 3 only
- C** 1 and 3 only
- D** 1 and 2 only ✓

#Q. The simplified output of $Y = AB + \overline{AC} + \underline{A\overline{B}C}(AB + C)$ is

- ☐ A 0
- ☐ B B
- ☐ C AB
- ☒ D None of these

$$Y = AB + \overline{AC} + A\overline{B}C(C + AB)$$

$$= AB + \overline{AC} + A\overline{B}C$$

$$= A[B + (\overline{B} \cdot C)] + \overline{AC}$$

$$= A[B + C] + \overline{AC}$$

$$= AB + AC + \overline{AC}$$

$$= AB + \overbrace{AC + \overline{AC}}^1$$

$$= 1$$

#Q. The minimum number of 2 input NAND gates required to realize the equation $y = \bar{A}(B + C + D)$ is,

A 10

B 9

C 8

D 7

$$Y = \bar{A}B + \bar{A}C + \bar{A}D$$

$$\bar{Y} = \overline{\bar{A}B} \cdot \overline{\bar{A}C} \cdot \overline{\bar{A}D} = P \cdot Q \cdot R$$

$\downarrow \quad \quad \downarrow \quad \quad \downarrow$
 2NAND 1NAND 1NAND

$$Y = \overline{P \cdot Q \cdot R} \longrightarrow 3 \text{ NAND}$$

(7)

#Q. Consider the following K-MAP, The no. of PI and EPI will be,

- A** 8 (PI) , 5 (EPI)
- B** 5 (PI), 4 (EPI)
- C** 4 (PI), 4 (EPI)
- D** 5 (PI), 5 (EPI)

wx \ yz	yz			
	00	01	11	10
00			1	
01	1	1	1	
11		1	1	1
10		1		

PI $\rightarrow$ 5 $\rightarrow$ 4 (EPI)

#Q. Consider the Karnaugh map given below: The function represented by this map can be simplified to the minimal form as

- A** $x_1 \bar{x}_2 x_4 + x_2 x_4 + x_1 \bar{x}_3$
- B** $x_1 x_2 x_4 + x_2 x_4 + x_1 \bar{x}_2 \bar{x}_3 \bar{x}_4$
- C** $x_2 x_4 + \bar{x}_2 \bar{x}_4 + x_1 \bar{x}_3$
- D** $x_1 \bar{x}_2 \bar{x}_4 + \bar{x}_1 x_2 \bar{x}_3 x_4 + x_1 x_2$

		$x_1 x_2$			
		00	01	11	10
$x_3 x_4$	00	1		d	d
	01		1	d	1
	11		d	1	
	10	1	d		d

#Q. The K-map for a Boolean function is shown in figure. The number of essential prime implicants for this function is

- A** 4 ✓
- B** 5 ✗
- C** 6 ✗
- D** 8 ✗

		AB			
CD		00	01	11	10
		1	1	0	1
01		0	0	0	1
11		1	0	0	0
10		1	0	0	1

PI - 4
EPI - 4

#Q. If the function X, Y are $X = R + P'Q + R'S$ and $Y = P'Q'R'S' + P'Q'R'S + PQR'S'$ (where ' represent complement). Which of the following is true?

- ☐ A $X = Y$
- ☒ B $X = Y'$
- ☐ C no direct relation between X and Y
- ☐ D insufficient data

$$X = \sum (1, 2, 3, 4, 5, 6, 7, 9, 10, 11, 13, 14, 15)$$

$$Y = \sum (0, 8, 12)$$

PQ R. S	$\bar{P}Q$
0 0 1 0 $\rightarrow$ 2	0 1 0 0 $\rightarrow$ 4
0 0 1 1 $\rightarrow$ 3	0 1 0 1 $\rightarrow$ 5
0 1 1 0 $\rightarrow$ 6	0 1 1 0 $\rightarrow$ 6
0 1 1 1 $\rightarrow$ 7	0 1 1 1 $\rightarrow$ 7
1 0 1 0 $\rightarrow$ 10	
1 0 1 1 $\rightarrow$ 11	
1 1 1 0 $\rightarrow$ 14	
1 1 1 1 $\rightarrow$ 15	

$R'S$
0 0 0 1 $\rightarrow$ 1
0 1 0 1 $\rightarrow$ 5
1 0 0 1 $\rightarrow$ 9
1 1 0 1 $\rightarrow$ 13

#Q. How many variables do 16 squares eliminate in K-map?

- A** 7
- B** 4
- C** 1
- D** 11

$2^1 \rightarrow \text{Variable}$
 $2^2 \rightarrow \text{Variable}$
 $2^3 \rightarrow \text{Variables}$
 $2^4 \rightarrow 4\text{-Variables}$

#Q. Five panelists are required to elect a sixth member to the panel. If any of the panelists votes against a member, the member is disqualified. What would be the appropriate electronic circuit to be used in the electronic voting machine to implement the above rule?

- A** XOR
- B** XNOR
- C** OR
- D** AND

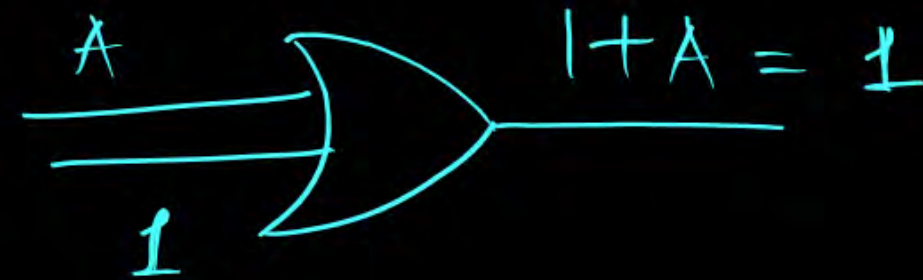
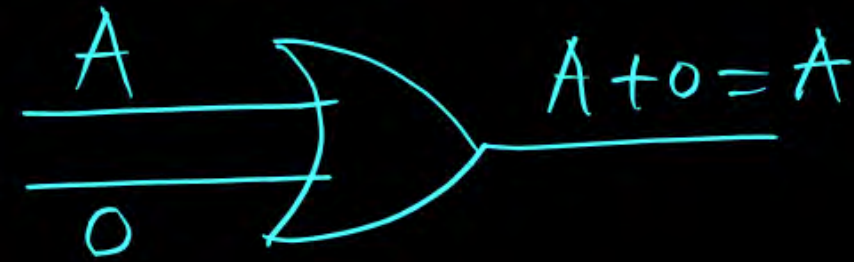


#Q. Consider the statements below:

- A. If the output waveform from an OR gate is the same as the waveform at one of its inputs, the other input is being held permanently LOW. ✓
- B. If the output waveform from an OR gate is always HIGH, one of its input is being held permanently HIGH. ✓

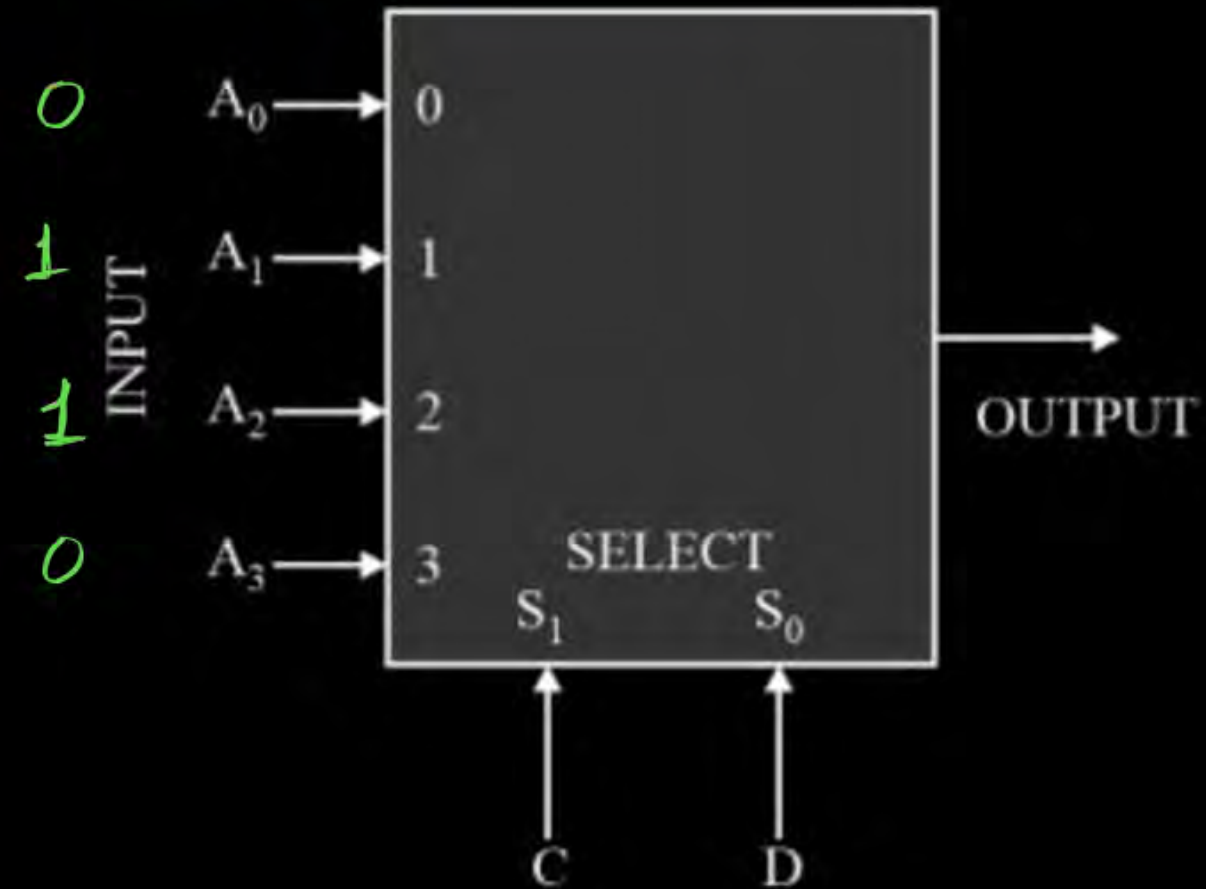
The statement, which is always true, is

- ☒ A Both A and B
- ☐ B Only A
- ☐ C Only B
- ☐ D None of the above



#Q. Consider the 2-bit multiplexer (MUX) shown in the figure. For OUTPUT to be the XOR of C and D, the values for A_0 , A_1 , A_2 and A_3 are _____.

- A** $A_0 = 0, A_1 = 0, A_2 = 1, A_3 = 1$
- B** $A_0 = 1, A_1 = 0, A_2 = 1, A_3 = 0$
- C** $A_0 = 0, A_1 = 1, A_2 = 1, A_3 = 0$
- D** $A_0 = 1, A_1 = 1, A_2 = 0, A_3 = 0$



#Q. Without any additional circuitry, an 8:1 MUX can be used to obtain

$$2^3:1$$

(C)

A some but not all Boolean functions of 3 variables ✗

B all functions of 3 variables but none of 4 variables ✗

C all functions of 3 variables and some but not all of 4 variables ✓✓

D all functions of 4 variables ✗

$2^n:1$ $\rightarrow$ n -variable $\rightarrow$ All function
 $(n+1)$ variable $\rightarrow$ some of the function
 $2^n:1 + 1\text{NOT} \rightarrow (n+1)$ variable $\rightarrow$ all function

#Q. The following switching functions are to be implemented using a decoder:

$$f_1 = \sum m(1, 2, 4, 8, 10, 14)$$

$$f_2 = \sum m(2, 5, 9, 11)$$

$$f_3 = \sum m(2, 4, 5, 6, 7)$$

The minimum configuration of the decoder should be

- A** 2 - to - 4 lines
- B** 3 - to - 8 lines
- C** 4 - to - 16 lines
- D** 5 - to - 32 lines

#Q. To add two m -bit numbers, the required number of half adders is

- ☒ **A** $2m - 1$
- ☐ **B** $2^m - 1$
- ☐ **C** $2m + 1$
- ☐ **D** $2m$

$$1 \text{ H.A} + (m-1) \text{ F.A}$$

$$1 \text{ H.A} + (m-1) [2 \text{ HA} + 1 \text{ OR}]$$

$$1 \text{ H.A} + (2m-2) \text{ H.A} + (m-1) \text{ OR gate}$$

$$(2m-1) \text{ HA} + (m-1) \text{ OR gate}$$

#Q. How many full adder(s) are required in a serial adder to perform 8-bit addition?

- ☐ A Two
- ☒ B One
- ☐ C Three
- ☐ D Eight

#Q. Match List-I with List-II as per the codes given below:

List-I		List-II	
A.	A digital circuit designed to keep track of a number of events	1.	Active low
B.	A group of flip flops used to store a binary number	2.	Negation
C.	A circuit that will invert a digital level	3.	Register
D.	An action occurs when the input is low	4.	Counter

A ✓ A→4, B→3, C→2, D→1

~~**B**~~ A→3, B→2, C→4, D→1

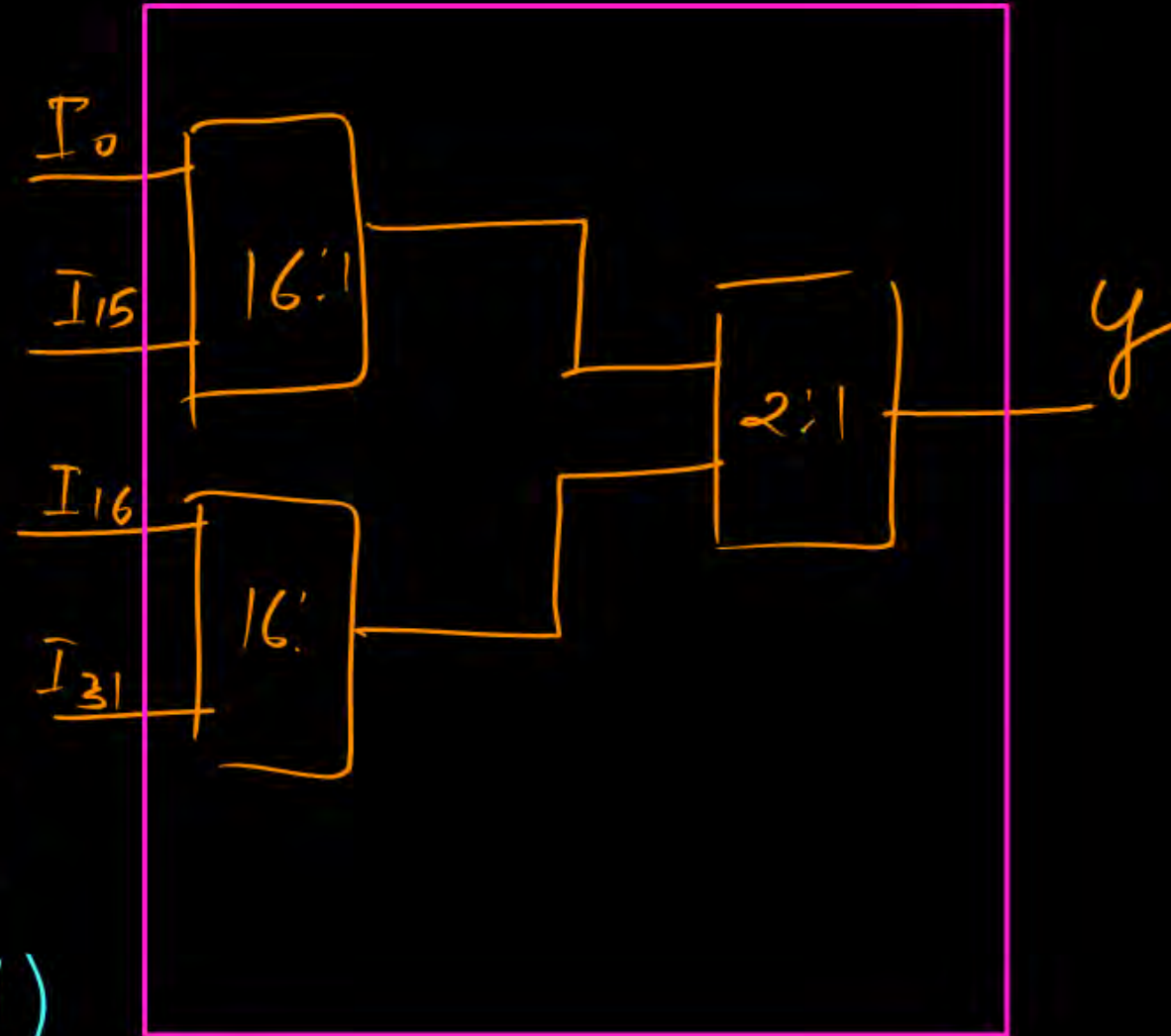
~~**C**~~ A→4, B→1, C→2, D→3

~~**D**~~ A→1, B→2, C→3, D→4

#Q. When two 16-input multiplexers drive a 2-input MUX, what is the result ?

- A** 2-input MUX
- B** 4-input MUX
- C** 16-input MUX
- D** 32-input MUX

$$32:1 \xrightarrow{16:1} 2(16:1) + 1(2:1)$$



#Q. The output Y of a 2-bit comparator is logic 1 whenever the 2-bit input A is greater than the 2-bit input B. The number of combinations for which the output is logic 1 is:

- A** 4
- B** 6
- C** 8
- D** 10

A	B
2 bit	2 bit
0	0
1	1
2	2
3	3

$$N_1(A > B) = N_2(A < B)$$

$$= 0 + 1 + 2 + 3$$

$$= 6$$

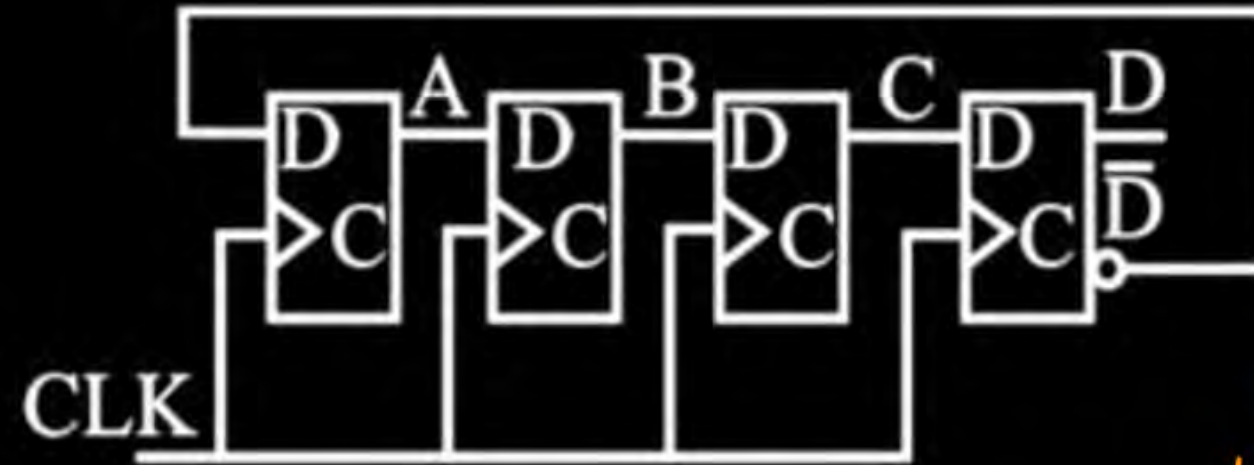
$$N_3(A = B) = 4$$

#Q. In the full adder, denoting sum by S and carry by C .

- A** $S = 1$ when two or more inputs are unity.
- B** $C = 1$ when two or more inputs are unity.
- C** $C = 1$ when all the inputs are unity. ✗
- D** $S = 1$ when all the inputs are unity.

#Q. If the following counter is initially at 0000 state then after 6th clock ABCD will be -

- A** 0111
- B** 0011
- C** 0001
- D** 1000



	A	B	C	D
Initial	0	0	0	0
1 st	1	0	0	0
2 nd	1	1	0	0
3 rd	1	1	1	0
4 th	1	1	1	1
5 th	0	1	1	1
6 th	0	0	1	1

#Q. If the input to T-flip-flop is 100 Hz signal, the final output of the three T-flip-flops that are connected in cascade is

- A** 1000 Hz ✗
- B** 500 Hz ✗
- C** 333 Hz ✗
- D** 12.5 Hz ✓

#Q. D input of a clocked D-flip-flop receives an input $A \oplus Q_n$ where A is an external logic input and Q_n is the output of the n^{th} D-FF before the clock appears. The circuit works as

- A** EX-OR gate
- B** T-FF
- C** D-FF
- D** JK-FF

$$Q(n+1) = D$$

$$Q(n+1) = A \oplus Q(n)$$

└──────────┘
 ↓
 T ff

#Q. A J-K Flip-Flop may be used as T- Flip-Flop by connecting,

- A** Outputs to inputs in inverting mode
- B** Both the inputs
- C** Both the inputs using a inverter
- D** Output to corresponding input

$$J = T$$

$$K = T$$

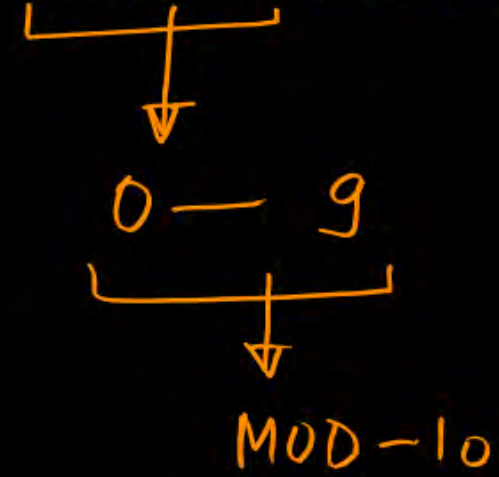
$$\begin{aligned} Q(n+1) &= J\bar{Q} + \bar{K}Q \\ &= T\bar{Q} + \bar{T}Q \\ &= T \oplus Q(n) \end{aligned}$$

$$J = D$$

$$K = \bar{D}$$

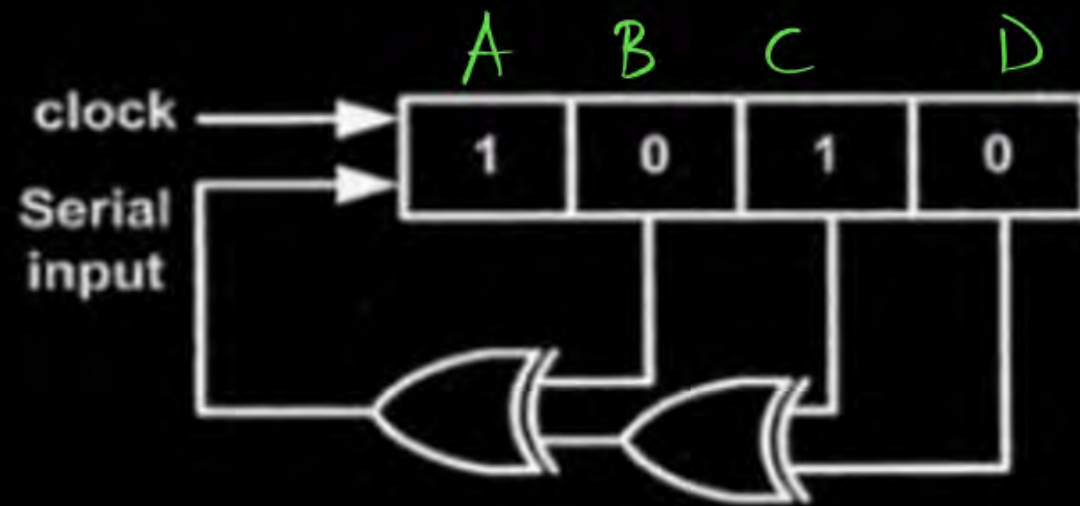
#Q. Minimum number of J-K flip-flops needed to construct a BCD counter is

- A** 2
- B** 3
- C** 4
- D** 5



#Q. Following shift register is initially loaded with the bit pattern "1010". After how many clock cycles will the content of shift register be "1010" again?

- A** 5
- B** 9
- C** 7
- D** 15



	1010
1 st	1101
2 nd	0110
3 rd	0011
4 th	0001
5 th	1000
6 th	0100
7 th	<u>1010</u>

#Q. A certain J-K flip-flop has propagation delay 12 picoseconds. What is the largest MOD of a counter that can be constructed from these J-K flip-flops and operates up to 10 GHz?

- A** 64
- B** 128
- C** 256
- D** 512

$$T_{clk} \geq n t_{df}$$

$$f_{clk} \leq \frac{1}{n t_{df}}$$

$$n \leq \frac{1}{f_{clk} t_{df}}$$

$$n \leq \frac{1}{10 \times 10^9 \times 12 \times 10^{-12}}$$

$$n \leq \frac{100}{12} \rightarrow n \leq 8.33$$

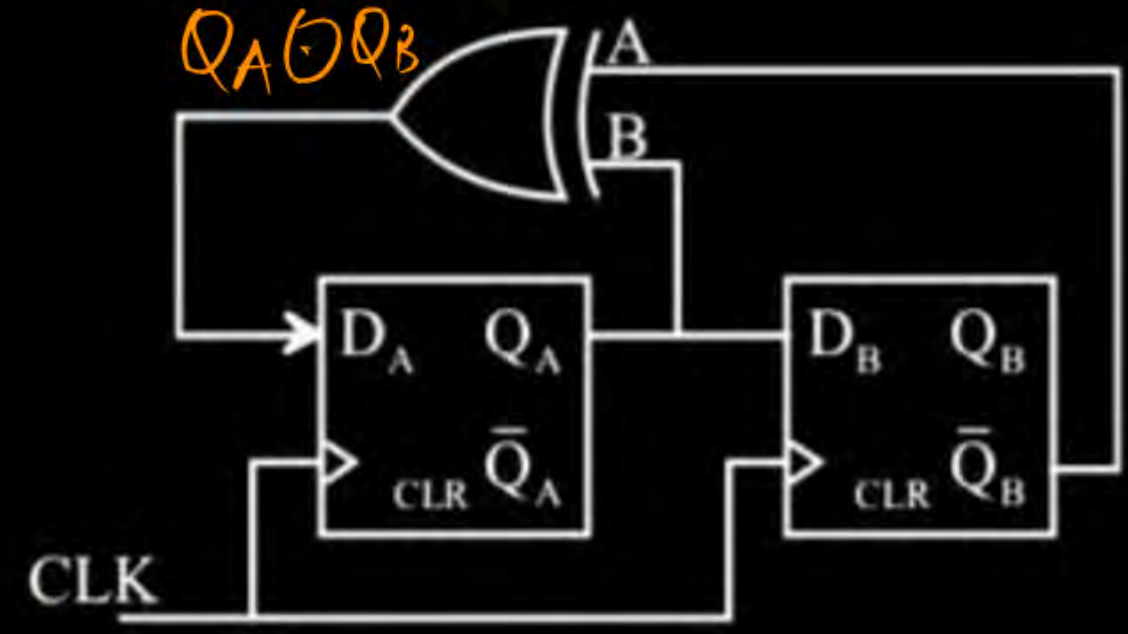
$$n \leq 8$$

$$n_{max} = 8$$

$$\Rightarrow M = 2^8 = 256$$

#Q. In the following circuit, what sequence is followed by A and B on rising edge of CLK after reset is de-asserted?

- A** ☒ AB = 10, 11, 00, 10, 11, ...
- B** ☒ AB = 10, 01, 00, 11, 10, ...
- C** ☒ AB = 10, 00, 01, 10, 00, ...
- D** ☒ AB = 11, 01, 00, 10, 11, ...



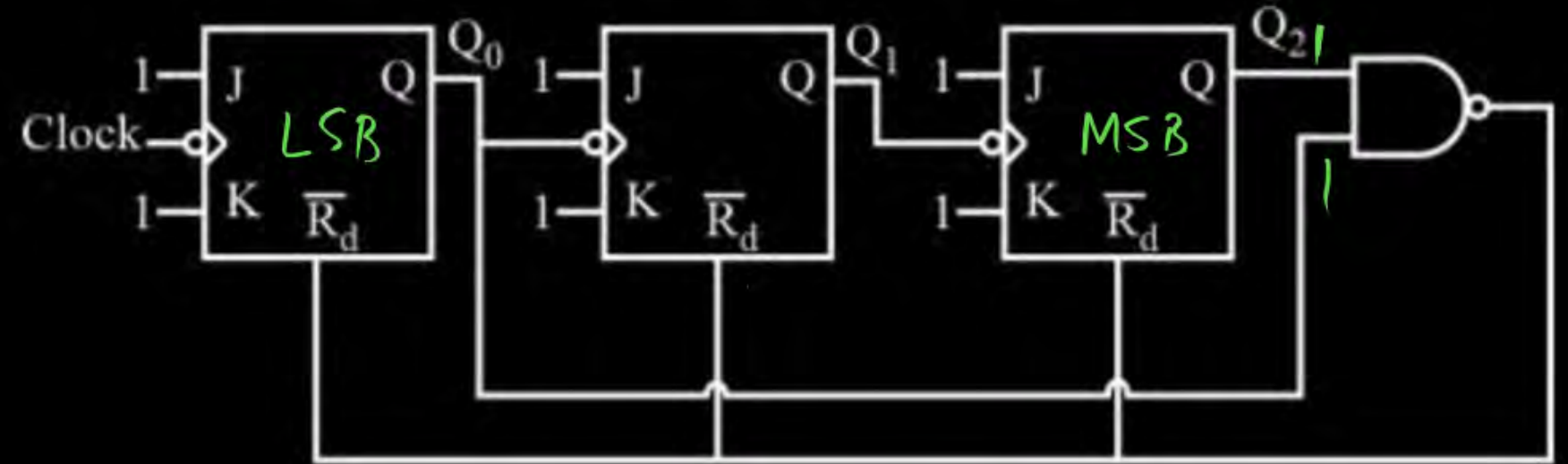
$$Q_A(n+1) = D_A = Q_A(n) \oplus Q_B(n)$$

$$Q_B(n+1) = D_B = Q_A(n)$$

	Q_A	Q_B
1 st	0	0
2 nd	1	0
	0	1

#Q. The circuit shown consists of J-K flip-flops, each with an active low asynchronous reset ($\bar{R}_D$ input). The counter corresponding to this circuit is

- A** ✓ A modulo-5 binary up counter
- B** A modulo-6 binary down counter ✗
- C** A modulo-5 down counter ✗
- D** A modulo-6 binary up counter



MOD - 5

up-counter

	Q_2	Q_1	Q_0
1 st	0	0	0
2 nd	0	0	1
3 rd	0	1	0
4 th	0	1	1
5 th	1	0	0
	1	0	1

A green arrow points from the 5th state (1 0 0) to the 1st state (0 0 0), indicating a modulo-5 up counter.

#Q. Which of the following technology results in least power dissipation?

- ☒ **A** CMOS
- ☐ **B** ECL
- ☐ **C** TTL
- ☐ **D** NMOS

#Q. Which of the following is 'true' for an ECL circuit ?

- I. Switching speed is highest among commercially available logic families ✓
- II. ECL logic block usually produces output and its complement also ✓
- III. Total current flow in an ECL remains relatively constant regardless of logic state
- IV. ECL also called Merged transistor logic I^2L ✗

A II, III and IV ✗

C I and IV ✗

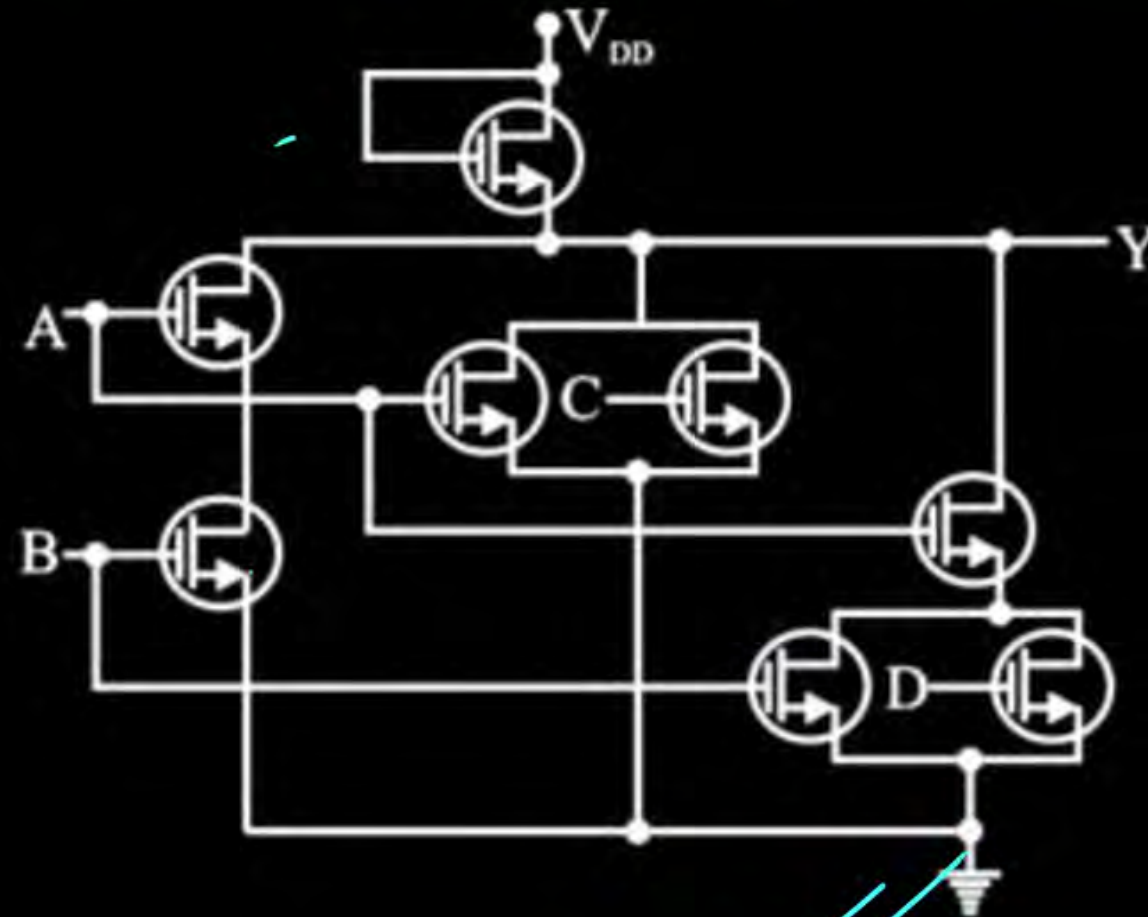
B I, II and III ✓

D II and IV ✗

#Q. In standard TTL, the 'totem pole' refers to

- A** Multi-emitter input stage ✗
- B** The phase splitter ✗
- C** Open collector output stage ✗
- D** The output buffer ✓✓

#Q. The logic function implemented by following circuit can be represented as



$$\left[A \cdot B + (A + C) + A(B + D) \right]$$

A $Y = [(AB) + (B+C) + (B+D)]'$

C $Y = [(AB)' + (A+C)' + (AB+D)]$

B $Y = [(AB) + (A+C) + A(B+D)]'$

D $Y = [(AB) + (A+C) + (AD)]'$

#Q. The conversion time for a 10-bit successive approximation A/D converter, for a clock frequency of 1 MHz is

$$T_c = nT_{clk} = 10T_{clk} = 10 \frac{1}{1\text{MHz}} = \frac{10}{10^6\text{Hz}}$$

- A** $1\mu\text{s}$
C $10\mu\text{s}$

- B** $5\mu\text{s}$
D $15\mu\text{s}$

$$= 10 \times 10^{-6} \text{ sec} \\ = 10\mu\text{sec}$$

#Q. The number of resistors required for an N bit DAC in the case of R-2R ladder is:



- A** N
- B** $2N$
- C** $3N$
- D** 2

$$\begin{array}{l} 2R \rightarrow (N+1) \\ R \rightarrow (N-1) \end{array} \rightarrow 2N$$

#Q. The number of comparators required for implementing an 8-bit flash analog-to-digital converter is

A 8
C 255

B 128
D 256

← $(2^n - 1)$ Comparators
 $2^n \rightarrow$ Resistances
255 - Comparators.

$2^n : n$ Priority Encoder

#Q. What value of X satisfies the equation $\sqrt{(121)_7} = (x)_2$?

(where integer numbers written as subscripts are the base of the corresponding number systems.)

A 1011

B 0111

C 1010

☒ **D** 1000

$$\begin{aligned} \overset{7^2}{1}\overset{7^1}{2}\overset{7^0}{1}_7 &= (7^2 \times 1 + 7^1 \times 2 + 7^0 \times 1)_{10} \\ &= (49 + 14 + 1)_{10} \\ &= (64)_{10} \end{aligned}$$

$$\sqrt{(64)_{10}} = (8)_{10} = (1000)_2$$

#Q. 11001, 1001 and 111001 correspond to the 2's complement representation of which one of the following sets of number ?

- A** 25, 9 and 57 ~~respectively~~
- B** - 6, - 6 and - 6 respectively
- C** -7, - 7 and - 7 respectively
- D** - 25, - ~~9~~ and - 57 respectively

$$\begin{aligned} & \frac{1001}{11001} = 11001 = 111001 \\ & 111000 \\ & \begin{pmatrix} -8 & 4 & 2 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}_2 = -8 + 0 + 0 + 1 \\ & = (-7)_{10} \end{aligned}$$



2 mins Summary

→ Question Practice



THANK
YOU